Индивидуальное задание

Задание № 14:

1) $\lim\limits_{n\to\infty} \dfrac{7n\sqrt[3]{n^2-2n}+n\sqrt{9n^2+7n-4}}{\sqrt[3]{n^3-2n^2+3}+2n\sqrt[4]{81n^4-18n^2+6}} = \lim\limits_{n\to\infty} \dfrac{7\sqrt[3]{\frac{1}{n}-\frac{2}{n^2}}+\sqrt{9+\frac{7}{n}-\frac{4}{n^2}}}{\sqrt[3]{1-\frac{2}{n}+\frac{3}{n^3}}+2\sqrt[4]{81-\frac{8}{n^2}+\frac{6}{n^4}}} =$

$= \lim\limits_{n\to\infty} \dfrac{0+3}{0+2\cdot 3} = \dfrac{1}{2}$

2) $\lim\limits_{n\to\infty} \dfrac{6^{n+3}-4^{n+2}}{2^{n-1}\cdot 3^{n+2}+3n^6} = \lim\limits_{n\to\infty} \dfrac{6-\left(\frac{2}{3}\right)^{n+2}}{\frac{1}{8}+\frac{3n^6}{6^{n+2}}} = \lim\limits_{n\to\infty} \dfrac{6-0}{\frac{1}{8}+0} = 48$

3) $\lim\limits_{n\to\infty} \dfrac{26^{n+3}+4(n-5)\cdot n!}{12^{n-1}-5\cdot(n+1)!} = \lim\limits_{n\to\infty} \dfrac{\frac{26^{n+3}}{(n+1)!}+4\left(1-\frac{5}{n}\right)}{\frac{12^{n-1}}{(n+1)!}-5} = \lim\limits_{n\to\infty} \dfrac{0+4}{0-5} = -\dfrac{4}{5}$

4) $\lim\limits_{n\to\infty} \left(\dfrac{2n^2-3}{6n^2+5}\right)^{3n} = \lim\limits_{n\to\infty} \left(\dfrac{2-\frac{3}{n^2}}{6+\frac{5}{n^2}}\right)^{3n} = \lim\limits_{n\to\infty} \left(\dfrac{1}{3}\right)^{3n} = 0$

5) $\lim\limits_{n\to\infty} \sqrt[n]{7^{n-2}+25n^5} = \lim\limits_{n\to\infty} 7\cdot\sqrt[n]{\frac{1}{49}+\frac{25n^5}{7^n}} = 7$

6) $\lim\limits_{n\to\infty} \left(\dfrac{2n^2-3}{2n^2+7}\right)^{3n^2} = \lim\limits_{n\to\infty} \left(1+\dfrac{-10}{2n^2+7}\right)^{3n^2} = \lim\limits_{n\to\infty} \left(\left(1+\dfrac{-10}{2n^2+7}\right)^{\frac{2n^2+7}{-10}}\right)^{3n^2\cdot\frac{-10}{2n^2+7}} =$

$= \lim\limits_{n\to\infty} e^{-15\cdot\left(\frac{2n^2}{2n^2+7}\right)} = \lim\limits_{n\to\infty} e^{-15\cdot\left(1-\frac{7}{2n^2+7}\right)} = \lim\limits_{n\to\infty} e^{-15\cdot 1} = e^{-15}$

7) $\lim\limits_{n\to\infty} \sqrt[n]{7^{n-2}\cdot n + 25n^5} = \lim\limits_{n\to\infty} \sqrt[n]{n}\cdot 7\cdot \sqrt[n]{\frac{1}{49} + \frac{25n^4}{7^n}} = 1\cdot 7\cdot 1 = 7$

8) $\lim\limits_{n\to\infty} \frac{\sqrt[n]{2}-1}{\sqrt[n]{4^7}-11\cdot\sqrt[n]{2}+10} = \lim\limits_{n\to\infty} \frac{(\sqrt[n]{2}-1)}{(\sqrt[n]{2}-10)\cdot(\sqrt[n]{2}-1)} = \frac{1}{1-10} = -\frac{1}{9}$

9) $\lim\limits_{n\to\infty} \frac{3}{n(\sqrt{4n^2+7}-\sqrt{4n^2-5})} = \lim\limits_{n\to\infty} \frac{3\cdot(\sqrt{4n^2+7}+\sqrt{4n^2-5})}{n\cdot 12} =$

$= \lim\limits_{n\to\infty} \frac{6n(\sqrt{1+\frac{7}{4n^2}}+\sqrt{1-\frac{5}{4n^2}})}{n+2} = \frac{2}{2} = 1$

10) $\lim\limits_{n\to\infty} \frac{1-5+9-13+\ldots+(4n+1)-(4n+5)}{8n-7} = \lim\limits_{n\to\infty} \frac{(1+9+17+\ldots)-(5+13+\ldots)}{8n-7} =$

$= \lim\limits_{n\to\infty} \frac{n(2n+1)-n(2n+3)}{8n-7} = \lim\limits_{n\to\infty} \frac{-2n}{8n-7} = -\frac{1}{4}$

Задание 5. N 14

$$x_n = \frac{4n+5}{3n-2} \cdot \arccos\frac{3+(-1)^n}{4}$$

$\cdot n = 2k$: $x_n = \frac{8k+5}{6k-2} \cdot \arccos 1 = 0$

$\cdot n = 2k-1$: $x_n = \frac{8k+1}{6k-5} \cdot \arccos\frac{1}{2} = \frac{\pi}{3} \cdot \frac{8k+1}{6k-5} = \frac{\pi}{3} \cdot \frac{4}{3} \cdot \frac{6k+\frac{3}{4}}{6k-5} =$

$= \frac{4\pi}{9}\left(1 + \frac{5\frac{3}{4}}{6k-5}\right) \le \frac{4\pi}{9} \cdot \frac{2\cdot 3}{4} = 3\pi \ (n=1) \Rightarrow \begin{vmatrix} \sup x_n = 3\pi \\ \inf x_n = 0 \end{vmatrix}$

$\varlimsup\limits_{n\to\infty} x_n = 0$; $\varliminf\limits_{n\to\infty} x_n = \frac{4\pi}{9}$

Задание 6. N 14

$x_n = x_{n-1} + \frac{(-1)^n}{6^n}$; $x_1 = a$; $x_n = a + \sum\limits_{k=2}^{n} \frac{(-1)^k}{6^k}$

Критерий Коши: $\forall \varepsilon > 0 \ \exists n_0 = n(\varepsilon) \in \mathbb{N} : \forall n > n_0 \ \forall p \in \mathbb{N}$

$|x_{n+p} - x_n| < \varepsilon$

(сх $\Leftrightarrow$ фунд)

$x_{n+p} = a + \sum\limits_{k=2}^{n+p} \frac{(-1)^k}{6^k} \Rightarrow |x_{n+p} - x_n| = \sum\limits_{k=n+1}^{n+p} \frac{(-1)^k}{6^k} < \sum\limits_{k=n+1}^{n+p} \frac{1}{6^k} =$

$= \frac{\frac{1}{6^{n+1}}\left(1-\frac{1}{6^p}\right)}{1-\frac{1}{6}} = \frac{1}{5\cdot 6^n}\left(1-\frac{1}{6^p}\right) < \frac{1}{6^n} < \varepsilon \Rightarrow 6^n > \frac{1}{\varepsilon} \Rightarrow n_0 = [\log_6 \frac{1}{\varepsilon}]$

$\forall \varepsilon > 0$ т.е. x_n - фундаментальна $\Rightarrow x_n$ сх-ся.

Задание 7. N 14(15)

$x_{n+1} = \frac{x_n + 1}{3}, \; x_1 = 1$

• $\frac{x_{n+1}}{x_n} = \frac{1}{3}\frac{x_n+1}{x_n} = \frac{1}{3}\cdot\left(1+\frac{1}{x_n}\right) \leq \frac{2}{3} \; (n=1) < 1 \Rightarrow x_n$ - убывает

$x_n > 0 \Rightarrow$ по т. Вейерштрасса x_n сх-ся.

• $x_{n+1} = \frac{x_n+1}{3} < \frac{x_n}{3}$

$\Delta \; y_{n+1} = \frac{y_n}{3}; \; y_1 = x_1 = 1 \Rightarrow y_{n+1} = \frac{1}{3^n}$; $\frac{1}{3^n} < \varepsilon \quad \forall \varepsilon > 0$

$y_{n+1} < x_{n+1} \; \forall n \in \mathbb{N} \Rightarrow$

$\Rightarrow x_n > y_n; \; y_n \xrightarrow[n\to\infty]{} 0$

$0 < x_n < y_n \Rightarrow$ по лемме о стаей пер-ой $x_n \xrightarrow[n\to\infty]{} 0$

$\lim\limits_{n\to\infty} x_n = 0.$

№14 $x_{n+1} = \frac{x_n + 3}{4}$, $x_1 = 2$

$\frac{x_{n+1}}{x_n} = \frac{1}{4}\left(1 + \frac{3}{x_n}\right) \le \frac{1}{4} \cdot \left(1 + \frac{3}{2}\right) = \frac{1}{4} \cdot \frac{5}{2} = \frac{5}{8} < 1 \Rightarrow$

$\Rightarrow x_n$ убывает; $x_n > 0$

по т. Вейерштрасса x_n сх-ся

$\triangle\ y_{n+1} = \frac{y_n}{4}$, $y_1 = x_1 = 2 \Rightarrow y_{n+1} = \frac{2}{4^n} < x_{n+1}$

$y_{n+1} \underset{n\to\infty}{\to} 0$, т.к. $\frac{2}{4^n} < \varepsilon$ $\forall \varepsilon > 0$

$0 < x_{n+1} < y_{n+1} \Rightarrow x_{n+1} \underset{n\to\infty}{\to} 0 \Rightarrow x_n \underset{n\to\infty}{\to} 0.$

$\lim\limits_{n\to\infty} x_n = 0$